Math 118: Mathematical Methods of Data Theory

Lecture 12: Introduction to Linear Programming

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Optimization

- Thus far, we have focused on building **models** for Data Science.
- Examples:
  - Solving Least Squares.
  - SVD and PCA.
  - k-means.
  - Spectral Clustering.
  - Non-negative matrix factorization.
- Key computational step: solve a **minimization problem** $\min_{x \in \mathcal{X}} f(x)$
- Examples:
  - Least Squares: $\min_{\boldsymbol{x} \in \mathbb{R}^m} \|A\boldsymbol{x} - \boldsymbol{b}\|_2$.
  - k-means: $\min_{\Pi \text{a partition vector}} Q(\Pi)$ where $Q(\Pi) = \sum_{a=1}^{k} \sum_{j:\Pi_j = a} \|\mu_a - \mathbf{x}_j\|_2^2$.
  - NMF: $\min_{W \in \mathbb{R}_{\geq 0}^{m \times r}, H \in \mathbb{R}_{\geq 0}^{r \times n}} \|X - WH\|_2$
- In the next few weeks we'll focus on theory and practice of **optimization**.

Outline of next few weeks

Standard Optimization Problem: $\min_{x \in \mathcal{X} \subset \mathbb{R}^m} f(x)$

- Linear Programming:
  - $f(x) = \boldsymbol{c}^\top \boldsymbol{x}$
  - $\mathcal{X}$ defined by linear inequalities.
- Convex Optimization:
  - $f(x)$ a *convex* function.
  - Unconstrained: $\mathcal{X} = \mathbb{R}^m$.
  - Constrained: $\mathcal{X} \subset \mathbb{R}^m$ and convex.
- Combinatorial Optimization.
  - Dynamic Programming.
  - Integer Programming.

Linear Programming: Example 1 [1]

- Suppose at least 4 units of protein required in a diet.
- Suppose there are two sources of protein: steak and peanut butter.
- Suppose 1 pound steak yields 2 units protein and 1 pound peanut butter yields 1 unit protein.
- Suppose 1 pound steak costs $\$3$ and 1 pound peanut butter costs $\$2$.
- Let x_1 (resp. x_2) be amount of steak (resp. peanut butter) consumed, in pounds.
- **Cost:** $3x_1 + 2x_2 = \begin{bmatrix} 3 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \boldsymbol{c}^\top \boldsymbol{x}$
- **Goal:** Minimize $\boldsymbol{c}^\top \boldsymbol{x}$ subject to $2x_1 + 1x_2 \geq 4$ and $x_1 \geq 0$ and $x_2 \geq 0$.

[1] From Strang's *Linear Algebra and Applications*

Linear Programming: Example 2 [2]

- In one day, GM makes x_1 Chevrolets, x_2 Buicks and x_3 Cadillacs.
- Suppose that GM receives $\$100$ profit for Chevrolet, $\$200$ for Buick and $\$300$ for Cadillac.
- Suppose the fuel efficiencies of Chevrolets, Buicks and Cadillacs are $20, 17$ and 14 miles per gallon respectively and that by law the average fuel efficiency of vehicles manufactured must be 18 miles per gallon.
- Suppose Chevrolets, Buicks and Cadillacs require 1 minute, 2 minutes and 3 minutes respectively to manufacture. There are 480 minutes in a working day.
- **Profit:** $100x_1 + 200x_2 + 300x_3 = \begin{bmatrix} 100 & 200 & 300 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \boldsymbol{c}^\top \boldsymbol{x}$.
- **Goal:** Maximize $\boldsymbol{c}^\top \boldsymbol{x}$ subject to $20x_1 + 17x_2 + 14x_3 \geq 18(x_1 + x_2 + x_3)$ and $x_1 + 2x_2 + 3x_3 \leq 480$ and $x_1, x_2, x_3 \geq 0$.

[2] From Strang's *Linear Algebra and Applications*

Anatomy of a Linear Program

- **Cost function:** $\boldsymbol{c}^\top \boldsymbol{x}$.
- **Constraints:**(*e.g.* $2x_1 + 1x_2 \geq 4$). Rewrite as $\boldsymbol{a}_i^\top \boldsymbol{x} \geq b_i$ (*e.g.* here $\boldsymbol{a}_i = [2, 1]$ and $b_i = 4$) for $i = 1, \ldots, m$.
- **Feasible Set:**
  $\mathcal{X} = \left\{ \boldsymbol{x} : \ \boldsymbol{a}_i^\top \boldsymbol{x} \geq b_i \text{ for } i = 1, \ldots, m \text{ and } x_j \geq 0 \text{ for } j = 1, \ldots, n \right\}$
- **Geometrically:** Each constraint describes a **half-plane**. Feasible set is a **polyhedron**[3]

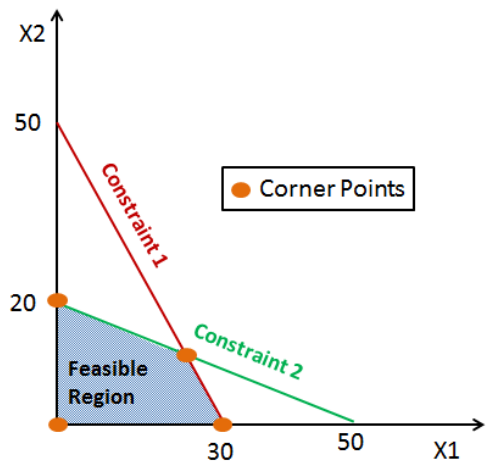


Figure: Taken from: `https://www.datacamp.com/community/tutorials/linear-programming-with-spreadsheets`

[3] By definition, a polyhedron is a finite intersection of half-planes. It is an example of a **convex** set

Anatomy of a Linear Program

- A point $\boldsymbol{x}$ is **feasible** if it is in the feasible region.
- **Corners** are corners of feasible regions.
- For any fixed value of cost function (*e.g.* 7), the condition $\boldsymbol{c}^\top \boldsymbol{x} = 7$ describes a straight line containing all points $\boldsymbol{x}$ with cost function equal 7.
- So the **minimizer** $\boldsymbol{x}^* = \underset{\boldsymbol{x} \in \mathcal{X}}{\arg\min}\, \boldsymbol{c}^\top \boldsymbol{x}$, if it exists, is a **corner of the feasible region**.

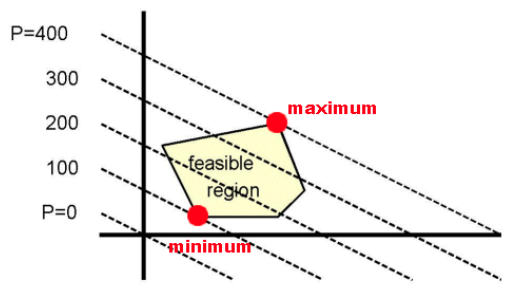


Figure: Taken from:
`http://orion.math.iastate.edu/mathbus/sp/s16/lessons/week09.html`

Idea of the Simplex Method

- If $\boldsymbol{x}^*$ is a solution to $\min_{\boldsymbol{x}\in\mathcal{X}} \boldsymbol{c}^\top \boldsymbol{x}$ then it must be a corner of $\mathcal{X}$.
- Solving Simple problems: find all corners $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_\ell$ of $\mathcal{X}$. Select one minimizing the cost: $\boldsymbol{x}_* = \arg\min_{i=1,\ldots,\ell} \boldsymbol{c}^\top \boldsymbol{x}_i$
- **The Simplex Method:** *(idea)*
  1. Find a corner $\boldsymbol{x}_1$.
  2. For $k = 1 : K_{\max}$
     - 2.1 Examine all corners adjacent to $\boldsymbol{x}_k$. Call them $\boldsymbol{z}_1, \ldots, \boldsymbol{z}_r$.
     - 2.2 Select one minimizing cost function: $\boldsymbol{z}_{j^*} = \arg\min_{j=1,\ldots,r} \boldsymbol{c}^\top \boldsymbol{z}_j$.
     - 2.3 If $\boldsymbol{c}^\top \boldsymbol{z}_{j^*} < \boldsymbol{c}^\top \boldsymbol{x}_k$ let $\boldsymbol{x}_{k+1} = \boldsymbol{z}_{j^*}$
     - 2.4 Else terminate and return $\boldsymbol{x}_* = \boldsymbol{x}_k$. This is the true minimizer.
  3. If $K_{\max}$ reached, return $\boldsymbol{x}_* = \boldsymbol{x}_{K_{\max}}$ and approximate solution.
- **Problem:** There can be many corners! For n variables and m constraints as many as $(n+m)!/n!m!$ corners!

Standardizing the constraints

- Typically write:

$$\min \boldsymbol{c}^\top \boldsymbol{x} \text{ subject to } \boldsymbol{a}_i^\top \boldsymbol{x} \geq b_i \text{ and } x_j \geq 0$$

  This is equivalent to:

$$\min_{\boldsymbol{x} \in \mathcal{X}} \boldsymbol{c}^\top \boldsymbol{x} \text{ where } \mathcal{X} = \left\{ \boldsymbol{x} \in \mathbb{R}^m : \ \boldsymbol{a}_i^\top \boldsymbol{x} \geq b_i \text{ and } x_j \geq 0 \right\}$$

- Currently have two kinds of constraints: $\boldsymbol{a}_i^\top \boldsymbol{x} \geq b_i$ and $x_j \geq 0$. Will reduce to just one kind using **slack variables**.
- **Slack Variables:** For every inequality $\boldsymbol{a}_i^\top \boldsymbol{x} \geq b_i$ (*e.g.* $x_1 + 2x_2 \geq 4$) introduce a new variable w_i (*e.g.* $w_i = x_1 + 2x_2 - 4$).
- Then: $\boldsymbol{a}_i^\top \boldsymbol{x} \geq b_i \Leftrightarrow w_i \geq 0$

Converting to Standard Form

- Consider: $\min 3x_1 + 5x_2$ subject to: $\begin{matrix} x_1 \geq 0 \\ x_2 \geq 0 \end{matrix}$ and $\begin{matrix} x_1 + 2x_2 \geq 4 \\ 2x_1 - 7x_2 \geq 2 \end{matrix}$
- Introduce **slack variables:** $w_1 = x_1 + 2x_2 - 4$ and $w_2 = 2x_1 - 7x_2 - 2$.
- Original problem becomes:

$$\min 3x_1 + 5x_2 \text{ subject to: } \begin{matrix} x_1 \geq 0 \\ x_2 \geq 0 \\ w_1 \geq 0 \\ w_2 \geq 0 \end{matrix} \text{ and } \begin{bmatrix} 1 & 2 & -1 & 0 \\ 2 & -7 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

- **Relabel:**

$$\begin{matrix} x_3 := w_1 \\ x_4 := w_2 \end{matrix} \quad A = \begin{bmatrix} 1 & 2 & -1 & 0 \\ 2 & -7 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix} \text{ and } \boldsymbol{b} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

- **Standard Form:** $\min \boldsymbol{c}^\top \boldsymbol{x}$ subject to: $\boldsymbol{x} \geq 0$ and $A\boldsymbol{x} = \boldsymbol{b}$
- Note that $\boldsymbol{c}^\top = [3 \quad 5 \quad 0 \quad 0]$

Summary

- Typical **Linear Program** (abbreviated to **LP**):

$$\min \boldsymbol{c}^\top \boldsymbol{x} \text{ subject to } \boldsymbol{a}_i^\top \boldsymbol{x} \geq b_i \; i = 1, \ldots, m \text{ and } x_j \geq 0 \; j = 1, \ldots, n$$

- **Feasible Region** $\mathcal{X} = \left\{ \boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{a}_i^\top \boldsymbol{x} \geq b_i \text{ and } x_j \geq 0 \right\}$ is a **polyhedron**.
- Solution to the LP must lie on a **corner** of $\mathcal{X}$.
- The **Simplex Method** systematically checks the corners of $\mathcal{X}$ to find a solution.
- Before applying simplex method first need to convert to **Standard Form**:

$$\min \boldsymbol{c}^\top \boldsymbol{x} \text{ subject to: } \boldsymbol{x} \geq 0 \text{ and } A\boldsymbol{x} = \boldsymbol{b}$$